

ULAGANATHAN A

QA Automation Engineer • SDET • Full Stack Developer (MERN)

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3+ Years

QA Automation
Healthcare & Fintech

40%

S3 Validation Reduction
Automation • Fintech

Philips

Key Contributor
Rel26H2 Reliability

PROFESSIONAL SUMMARY

QA Automation Engineer and SDET with **3+ years of experience** in **Healthcare (Philips, IEC 62304)** and **Fintech** domains. Skilled in **Python, C#, Behave BDD, SpecFlow, Selenium, REST API, and CI/CD pipelines**. Expert in **OCR validation (Tesseract)** and **OpenCV computer vision testing** — rare skills in regulated medical device QA. MERN-certified full stack developer. Currently learning **Playwright**.

TECHNICAL SKILLS

Languages: Python • C# • JavaScript • Java

Automation Frameworks: Behave (BDD) • SpecFlow (BDD) • Selenium WebDriver • Playwright (Learning) • Pytest • Cucumber / TestNG

Testing Types: API Testing • REST API • UI / E2E • Regression • Reliability • OCR Validation • Visual Regression

Computer Vision / AI: OpenCV (Template Matching) • Tesseract OCR • Entropy Image Quality • GitHub Copilot

Databases: SQL • MongoDB

CI/CD & DevOps: GitHub Actions • Jenkins • Azure DevOps • Docker • ControlHub Server

Cloud & Tools: AWS S3 • AWS EC2 • Postman • Swagger / OpenAPI • Jira • Git • Linux

Full Stack (MERN): React.js • Node.js • Express.js • MongoDB

PROFESSIONAL EXPERIENCE

Automation Test Engineer | **Randstad Enterprise Pvt Ltd (Client: Philips Healthcare)**

Jun 2025 – Present

Bengaluru • Healthcare / Medical Imaging • IVUS Clinical Imaging (IEC 62304 Regulated)

AUTOMATION & FRAMEWORK DESIGN

- Python + Behave BDD:** Built and maintained scalable E2E automation frameworks for Philips IVUS clinical imaging in an IEC 62304 regulated medical device environment.
- UI Automation:** Developed UI automation workflows using SendKey API and iMonkey device integration to simulate real-user clinical interactions.
- AI-Assisted Dev:** Leveraged GitHub Copilot to accelerate Python script authoring, debugging, and code review cycles.

VALIDATION & QUALITY ENGINEERING

- OpenCV Testing:** Implemented OpenCV template matching for dynamic UI element identification and image-driven visual validation on adaptive medical interfaces.
- Tesseract OCR:** Applied Tesseract OCR to verify clinical reports, annotations, and on-screen medical data — ensuring data integrity in IEC 62304 regulated workflows.
- Image Quality:** Introduced entropy-based image quality validation to ensure diagnostic-level accuracy of IVUS medical imaging outputs.
- API + Visual Regression:** Built API-driven screenshot capture pipelines to enforce UI consistency and visual regression detection across all builds.

CI/CD & RELIABILITY

- GitHub Actions CI/CD:** Configured GitHub Actions workflows and orchestrated execution via ControlHub server for centralized, scalable test management.
- Reliability Testing:** Conducted reliability, stability, and regression testing for life-critical clinical usage; performed root-cause analysis with dev teams.

Key Achievements

- **Philips Rel26H2 Key Contributor:** Recognised as a key contributor in enterprise-scale reliability testing initiative for life-critical IVUS imaging software.
- Significantly reduced manual verification effort by implementing **OCR, OpenCV, and API-based validation** — increasing automated test coverage across regulated clinical workflows.

Test Analyst | **Art Technology and Software Pvt Ltd**

Aug 2022 – Jul 2024

Kochi • Banking & Financial Services • Payment Gateway & Data Lake (Thredd)

AUTOMATION & API TESTING

- **Python + C# + SpecFlow:** Designed regression automation using Python, C#, and SpecFlow (BDD) with Selenium WebDriver for payment gateway and Thredd data lake platform.
- **BDD Framework (SpecFlow):** Implemented SpecFlow (C#) with Gherkin syntax to convert business requirements into executable test scenarios.
- **API Validation:** Executed REST API and data flow validation for chargeback, authorization, reporting, and reconciliation modules.

DATA & CLOUD VALIDATION

- **SQL + MongoDB:** Performed advanced database validation ensuring consistency and integrity of critical financial data across distributed systems.
- **AWS S3 Automation:** Built Python and C# with SpecFlow validation frameworks for AWS S3 data lake — automating large-scale financial dataset integrity checks.
- **Jenkins CI/CD:** Integrated test suites with Jenkins pipelines enabling continuous testing and faster deployment feedback loops.

Key Achievements

- ▶ **Reduced AWS S3 validation effort by 40%** through automation, improving data integrity assurance for the data lake platform.
- ▶ Contributed to **stable, on-time releases** across payment processing and data lake modules through automated regression workflows.

TECHNICAL PROJECTS

OCR Validation Framework • Python, Tesseract, GitHub Actions

Philips Healthcare

- Built Python automation framework using Tesseract OCR to validate clinical report text, medical annotations, and on-screen data in Philips IVUS imaging.
- Automated text extraction and comparison workflows — eliminating manual verification of clinical data in IEC 62304 regulated environment.

OpenCV Template Matching Framework • Python, OpenCV, Image Processing

Philips Healthcare

- Developed image comparison utilities using OpenCV to automate visual verification of IVUS medical imaging screens — replacing manual visual inspection.
- Implemented entropy-based image quality checks validating diagnostic accuracy of imaging outputs across adaptive medical interfaces.

Playwright Automation Framework • Python, Playwright, Page Object Model

Personal / Learning

- Building cross-browser E2E automation framework using Playwright (Python) with POM design pattern, HTML reporting, and GitHub Actions CI execution.

FULL STACK EXPOSURE (MERN)

MERN Stack Certified — React.js, Node.js, Express.js, MongoDB, REST API development. Full stack perspective strengthens test architecture decisions and dev-QA collaboration.

EDUCATION & CERTIFICATIONS

B.E. – Mechanical Engineering • Dhanalakshmi Srinivasan Engineering College

2020

- **Python Programming Certification** – GUVI (ID: 069B097t163g23MD5E)
- **Selenium Automation with C# Certification** – GUVI (ID: 069B097t163g23MD5E)
- **MERN Full Stack Development Certification** – GUVI (ID: 483cLnatC287072Ff1)

CURRENTLY LEARNING

- **Playwright (Python)** – Cross-browser E2E automation, Page Object Model, GitHub Actions integration
- **Advanced API Testing** – Contract testing, performance benchmarking, security validation patterns